

Diagonal Harmonics and Shuffle Theorems

George H. Seelinger

ghs9ae@virginia.edu

joint with Jonah Blasiak, Mark Haiman, Jennifer Morse, and Anna Pun

UVA Graduate Seminar

29 March 2021

- 1 Symmetric functions, S_n -representations, and Frobenius characteristic
- 2 Diagonal harmonics and shuffle conjectures
- 3 Stable series approach
- 4 Application: extended Delta conjecture

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Based off of slides from

- Mark Haiman: “A Shuffle Theorem for Paths Under Any Line”
<https://www.math.uwaterloo.ca/~opecheni/2020-06-12-AlCoVE.pdf>
- Jennifer Morse: “Hey Series, Tell Me About the Extended Delta Conjecture” (ICERM, March 22, 2021)

Multivariate Polynomials

- $f \in \mathbb{Q}[x_1, \dots, x_n]$ multivariate polynomial

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- $\sigma \in S_n$ acts as $\sigma.f(x_1, x_2, \dots, x_n) = f(x_{\sigma(1)}, x_{\sigma(2)}, \dots, x_{\sigma(n)})$

$$\begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} (5x_1^2 + 5x_2^2 + 8x_3^2) = 8x_1^2 + 5x_2^2 + 5x_3^2$$

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$$e_2 = x_1x_2 + x_1x_3 + x_2x_3 \quad h_2 = x_1^2 + x_1x_2 + x_1x_3 + x_2^2 + x_2x_3 + x_3^2$$

$$e_3 = x_1x_2x_3 \quad h_3 = x_1^3 + x_1^2x_2 + x_1^2x_3 + x_1x_2^2 + \dots$$

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$$e_3 = x_1x_2x_3 \quad h_3 = x_1^3 + x_1^2x_2 + x_1^2x_3 + x_1x_2^2 + \dots$$

- $\{f \in \mathbb{Q}[x_1, \dots, x_n] \mid \sigma.f = f \forall \sigma \in S_n\}$ forms a vector space, $\Lambda_{\mathbb{Q}}$.

Combinatorics of Symmetric Polynomials

Generators

$$e_r = \sum_{i_1 < i_2 < \dots < i_r} x_{i_1} x_{i_2} \cdots x_{i_r} \text{ or } h_r = \sum_{i_1 \leq i_2 \leq \dots \leq i_r} x_{i_1} x_{i_2} \cdots x_{i_r}$$

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Symmetric functions are polynomials in the $e_1, e_2, \dots$, or in the $h_1, h_2, \dots$

$$3h_2h_1^2 - h_2^2 + 6h_3h_1 = 3h_{(211)} - h_{(22)} + 6h_{(31)}$$

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Basis of $\Lambda_{\mathbb{Q}}$?

Definition

$n \in \mathbb{Z}_{>0}$, a *partition of n* is $\lambda = (\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_\ell > 0)$ such that $\lambda_1 + \lambda_2 + \cdots + \lambda_\ell = n$.

Partitions

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5 $\rightarrow$

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$4 + 1 \rightarrow$

$3 + 2 \rightarrow$

$3 + 1 + 1 \rightarrow$ 

$2 + 2 + 1 \rightarrow$

$$2 + 1 + 1 + 1 \rightarrow \begin{array}{|c|c|} \hline & \\ \hline & \\ \hline & \\ \hline & \\ \hline \end{array}$$

$$1 + 1 + 1 + 1 + 1 \rightarrow \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \square \\ \hline \square \\ \hline \square \\ \hline \end{array}$$

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Collection is called $\text{SSYT}(\lambda)$.

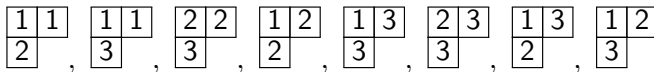
Definition

Filling of partition diagram of λ with numbers such that

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For $\lambda = (2, 1)$,

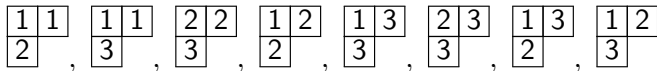


Schur functions

Associate a polynomial to $\text{SSYT}(\lambda)$.

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1	1	1	1	2	2	1	2	1	3	2	3	1	3	1	2
2		3		3		2		3		3		2		3	

$$s_{(21)}(x_1, x_2, x_3) = x_1^2 x_2 + x_1^2 x_3 + x_2^2 x_3 + x_1 x_2^2 + x_1 x_3^2 + x_2 x_3^2 + 2x_1 x_2 x_3$$

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$$\begin{array}{|c|c|} \hline 1 & 1 \\ \hline 2 & \\ \hline \end{array}, \begin{array}{|c|c|} \hline 1 & 1 \\ \hline 3 & \\ \hline \end{array}, \begin{array}{|c|c|} \hline 2 & 2 \\ \hline 3 & \\ \hline \end{array}, \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 2 & \\ \hline \end{array}, \begin{array}{|c|c|} \hline 1 & 3 \\ \hline 3 & \\ \hline \end{array}, \begin{array}{|c|c|} \hline 2 & 3 \\ \hline 3 & \\ \hline \end{array}, \begin{array}{|c|c|} \hline 1 & 3 \\ \hline 2 & \\ \hline \end{array}, \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 & \\ \hline \end{array}$$

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Definition

For λ a partition

$$s_\lambda = \sum_{T \in \text{SSYT}(\lambda)} x^T \text{ for } x^T = \prod_{i \in T} x_i$$

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- s_λ is a symmetric function
- Schur functions form a basis for $\Lambda_{\mathbb{Q}}$

Harmonic polynomials

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M = polynomials killed by all symmetric differential operators.

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Explicitly, for

$$\Delta = \det \begin{vmatrix} x_1^2 & x_1 & 1 \\ x_2^2 & x_2 & 1 \\ x_3^2 & x_3 & 1 \end{vmatrix} = x_1^2(x_2 - x_3) - x_2^2(x_1 - x_3) + x_3^2(x_1 - x_2)$$

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M is the vector space given by

$$\begin{aligned} M &= \text{sp} \left\{ \left(\partial_{x_1}^a \partial_{x_2}^b \partial_{x_3}^c \right) \Delta \mid a, b, c \geq 0 \right\} \\ &= \text{sp} \{ \Delta, 2x_1(x_2 - x_3) - x_2^2 + x_3^2, 2x_2(x_3 - x_1) - x_3^2 + x_1^2, \\ &\quad x_3 - x_1, x_2 - x_3, 1 \} \end{aligned}$$

Harmonic polynomials

$$\text{sp}\{\Delta, 2x_1(x_2 - x_3) - x_2^2 + x_3^2, 2x_2(x_3 - x_1) - x_3^2 + x_1^2, x_3 - x_1, x_2 - x_3, 1\}$$

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- 1 Break M up into irreducible S_n -representations.

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$$e_1^3 = (x_1 + x_2 + x_3)^3 = s_{\begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \square \\ \hline \end{array}} + s_{\begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \\ \hline \end{array}} + s_{\begin{array}{|c|c|} \hline \square & \square \\ \hline & \square \\ \hline \end{array}} + s_{\begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \end{array}}$$

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Schur basis expansion counts multiplicity of irreducible S_n -representations!

Upshot

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- 1 Schur functions $\leftrightarrow$ irreducible S_n -representations.

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- 2 Via Frobenius characteristic map, questions about S_n -action on vector spaces get translated to questions about Schur expansion coefficients in symmetric functions.

Getting more information

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Break M up into smallest S_n fixed subspaces

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Solution: irreducible S_n -representation of polynomials of degree $d \mapsto q^d s_\lambda$
(graded Frobenius)

$$?? = q^3 s_{\begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \square \\ \hline \end{array}} + q^2 s_{\begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \\ \hline \end{array}} + q s_{\begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \\ \hline \end{array}} + s_{\begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \end{array}}$$

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An example of bi-degree

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- $\mathbb{Q}[x_1, \dots, x_n, y_1, \dots, y_n]$ satisfying $\sigma(x_i) = x_{\sigma(i)}$, $\sigma(y_j) = y_{\sigma(j)}$.

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$$\Delta_{\begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \\ \hline \end{array}} = \det \begin{vmatrix} 1 & y_1 & x_1 \\ 1 & y_2 & x_2 \\ 1 & y_3 & x_3 \end{vmatrix} = x_3 y_2 - y_3 x_2 - y_1 x_3 + y_1 x_2 + y_3 x_1 - y_2 x_1$$

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An example of bi-degree

Capturing even more information...

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$$\tilde{H}_\mu = qts \begin{smallmatrix} \square \\ \square \\ \square \end{smallmatrix} + ts \begin{smallmatrix} \square & \square \\ \square \end{smallmatrix} + qs \begin{smallmatrix} \square & \square \\ \square \end{smallmatrix} + s \begin{smallmatrix} \square & \square & \square \end{smallmatrix}$$

- ① Symmetric functions, S_n -representations, and Frobenius characteristic
- ② **Diagonal harmonics and shuffle conjectures**
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- ④ Application: extended Delta conjecture

- $DH_n = \left\{ f \in \mathbb{Q}[x_1, \dots, x_n, y_1, \dots, y_n] \mid \sum_{1 \leq j \leq n} \partial_{x_j}^a \partial_{y_j}^b f(x, y) = 0 \right\}.$

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- E.g., Frobenius characteristic for DH_3 :

$$(q^3 + q^2t + qt^2 + t^3 + qt)s_{\begin{smallmatrix} \square \\ \square \\ \square \end{smallmatrix}} + (q^2 + qt + t^2 + q + t)s_{\begin{smallmatrix} \square & \square \\ \square \end{smallmatrix}} + s_{\square \square \square}$$

Diagonal harmonics

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Question

What symmetric function gives the Frobenius characteristic of DH_n ?

Frobenius characteristic of DH_3 :

Diagonal Harmonics

Frobenius characteristic of DH_3 :

$$\frac{t^3 \tilde{H}_{111}}{-qt^2 + t^3 + q^2 - qt} + \frac{(-q^2t - qt^2 - qt) \tilde{H}_{21}}{-q^2t^2 + q^3 + t^3 - qt} + \frac{-q^3 \tilde{H}_3}{-q^3 + q^2t + qt - t^2}$$

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Definition

Define $\nabla: \Lambda \rightarrow \Lambda$ via

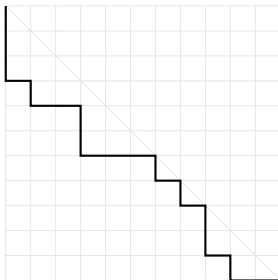
$$\nabla(\tilde{H}_\mu) = q^{n(\mu)} t^{n(\mu')} \tilde{H}_\mu$$

Nice, but not combinatorial...

Dyck paths

Dyck paths

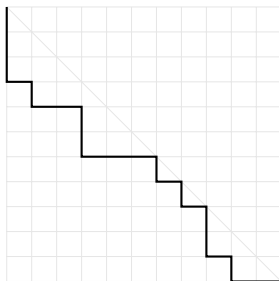
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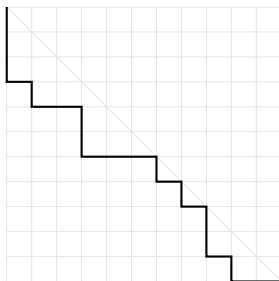


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- $\text{area}(\lambda)$ = number of squares above λ but below the path δ of alternating S-E steps.
- E.g., above $\text{area}(\lambda) = 10$.

Shuffle Conjecture

Conjecture (Haglund-Haiman-Loehr-Remmel-Ulyanov, 2005)

$$\nabla e_n = \sum_{\lambda \in \mathbf{DP}_n} t^{\text{area}(\lambda)} q^{\text{dinv}(\lambda)} \omega \mathcal{G}_{\nu(\lambda)}(x; q^{-1}).$$

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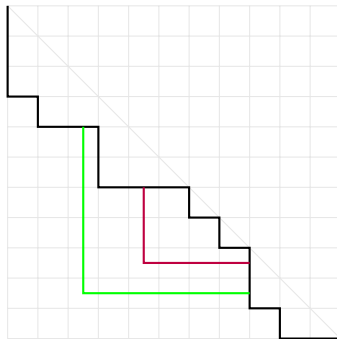
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- $\text{dinv}(\lambda) =$ number of balanced hooks.

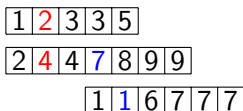


Balanced hook is given by a cell below λ satisfying

$$\frac{\ell}{a+1} < 1 - \epsilon < \frac{\ell+1}{a}.$$

$$\mathcal{G}_\nu(x; q^{-1}) = \sum_{T \in \text{SSYT}(\nu)} q^{-i(T)} x^T$$

for $i(T)$ the number of attacking inversions:



- $\mathcal{G}_\nu$ is symmetric and Schur positive.

Representation Theory: Diagonal Harmonics

$$DH_n = \left\{ f \in \mathbb{Q}[x_1, \dots, x_n, y_1, \dots, y_n] \mid \sum_{1 \leq j \leq n} \partial_{x_j}^a \partial_{y_j}^b f(x, y) = 0 \right\}$$

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Frobenius characteristic ∇e_n .

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Combinatorics: Shuffle Theorem (Carlsson-Mellit, 2018)

$$\nabla e_n = \sum_{\lambda \in \mathbf{DP}_n} t^{\text{area}(\lambda)} q^{\text{dinv}(\lambda)} \omega \mathcal{G}_{\nu(\lambda)}(x; q).$$

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Schiffmann's Elliptic Hall Algebra $\mathcal{E}$

- For every coprime $m, n \in \mathbb{Z}$, subalgebra $\Lambda(X^{m,n}) \cong \Lambda$, with relations between them. (Burban-Schiffmann, 2012)

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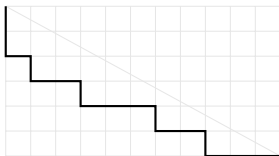
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Rational Shuffle Conjecture (F. Bergeron, Garsia, Sergel Leven, Xin, 2016) (Proved by Mellit, 2016)

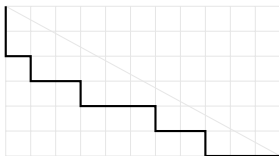
$$e_k[-MX^{m,n}] \cdot 1 = \sum_{\lambda} t^{\text{area}(\lambda)} q^{\text{dinv}_p(\lambda)} \omega_{\mathcal{G}_{\nu(\lambda)}}(X; q^{-1})$$

where summation is over all (kn, km) -Dyck paths.

Rational Path Combinatorics

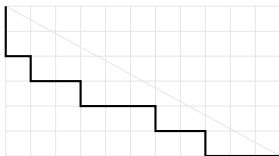


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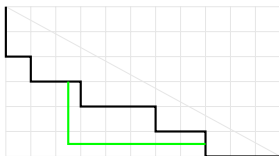


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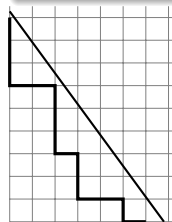
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Key Relationship

$$\omega(D_{\mathbf{b}} \cdot 1)(x_1, \dots, x_l) = H_{q,t} \left(\frac{x_1^{b_1} \dots x_l^{b_l}}{\prod_{i=1}^{l-1} (1 - qtx_i/x_{i+1})} \right)_{\text{pol}}$$

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For $\mathbf{b} \in \mathbb{Z}^l$ corresponding to some choice of coprime m, n ,

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- Under polynomial truncation, $\mathcal{L}_{\beta/\alpha}^\sigma \rightarrow q^{\text{dinv}_p(\lambda)} \mathcal{G}_{\nu(\lambda)}$

- (Twisted) non-symmetric Hall-Littlewood polynomials $E_{\lambda}^{\sigma}(x; q)$ defined via Demazure-Lusztig operators.

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- $\mathcal{L}_{\beta/\alpha} = H_q(w_0(F_{\beta}^{\sigma^{-1}}(x; q) \overline{E_{\alpha}^{\sigma^{-1}}(x; q)}))$

What have we learned?

Shuffle Theorem for any path

$$D_{\mathbf{b}} \cdot 1 = \sum_{\lambda} t^{\text{area}(\lambda)} q^{\text{dinv}_p(\lambda)} \omega \mathcal{G}_{\nu(\lambda)}$$

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Stable Shuffle Theorem

$$\begin{aligned} H_q \left(x^{\mathbf{b}} \frac{\prod_{i+1 < j} (1 - qtx_i/x_j)}{\prod_{i < j} (1 - tx_i/x_j)} \right) \\ = \sum_{a_1, \dots, a_{l-1} \geq 0} t^{|\mathbf{a}|} \mathcal{L}_{((b_l, \dots, b_1) + (0, a_{l-1}, \dots, a_1)) / (a_{l-1}, \dots, a_1, 0)}^{\sigma}(x; q) \end{aligned}$$

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- ① Symmetric functions, S_n -representations, and Frobenius characteristic
- ② Diagonal harmonics and shuffle conjectures
- ③ Stable series approach
- ④ **Application: extended Delta conjecture**

Another family of symmetric function operators

Changing the eigenvalues of Macdonald polynomials:

$$\Delta_f H_\mu = f[B_\mu] H_\mu \quad \Delta'_f H_\mu = f[B_\mu - 1] H_\mu$$

for any $f \in \Lambda$ and $B_\mu = \sum_{(i,j) \in \mu} q^{i-1} t^{j-1}$. (Note $\Delta'_{e_{n-1}} e_n = \nabla e_n$).

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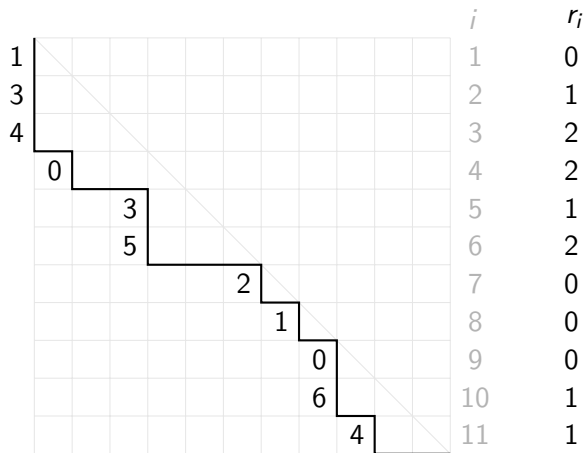
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Extended Delta Conjecture (Haglund-Remmel-Wilson, 2018)

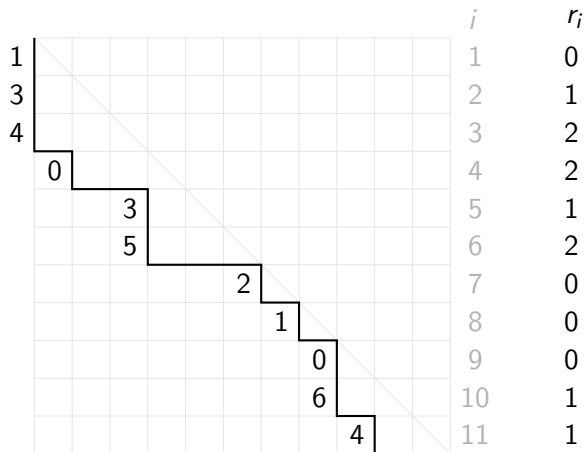
$$\Delta_{h_l} \Delta'_{e_{k-1}} e_n = \langle z^{n-k} \rangle \sum_{\lambda \in \mathbf{DP}_{n+l}} \sum_{P \in LD_{n+l,l}(\lambda)} q^{\text{dinv}(P)} t^{\text{area}(\lambda)} x^{\text{wt}_+(P)} \prod_{r_i(\lambda)=r_{i-1}(\lambda)+1} \left(1 + z t^{-r_i(\lambda)}\right)$$

Delta Combinatorics



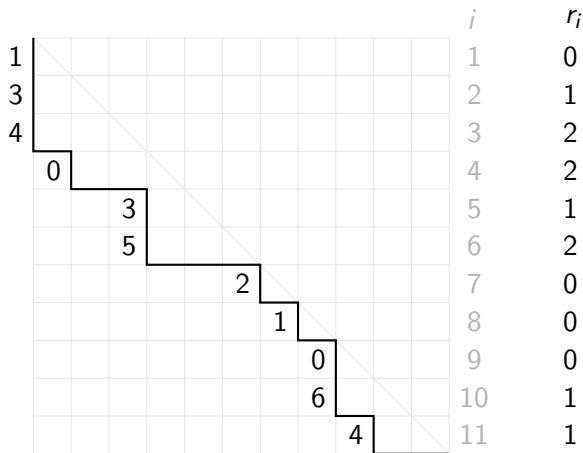
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Application of previous program

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 \end{aligned}$$

Stable Extended Delta Theorem

$$\begin{aligned}
 & H_q \left(\frac{\prod_{i+1 \leq j} (1 - qtx_i/x_j)}{\prod_{i < j} (1 - tx_i/x_j)} (x_1 \cdots x_{k+l}) h_{n-k}(x_1, \dots, x_{k+l}) \overline{e_l(x_2, \dots, x_{k+l})} \right) \\
 &= \sum_{\substack{J \subseteq [k+l-1] \\ |J|=l}} \sum_{\substack{(0, \mathbf{a}), \tau \in \mathbb{N}^{k+l} \\ |\tau|=n-k}} t^{|\mathbf{a}|} q^{d(\mathbf{a}, \tau, J)} \mathcal{L}_{\beta/\alpha}^{w_0}
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References

Thank you!

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